



Elastic Collisions in Two Dimensions

• Worksheet 1

Further Mechanics 1 Chapter 5: Elastic Collisions in Two Dimensions. Answer all questions. Show your working clearly.
Total: 47 marks. Time: 60 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are two sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ m s}^{-2}$. All spheres and surfaces are smooth: in an impact, only the velocity components along the common normal (the line of centres) change. Give angles in degrees and non-exact answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Oblique impact with a fixed surface	21	
B Oblique impact of two smooth spheres	21	
Roar-level challenge	5	
Total	47	

Section A Oblique impact with a fixed surface

21 marks

1. A smooth sphere strikes a smooth fixed vertical wall. Immediately before the impact, its velocity has a component of 4 m s^{-1} parallel to the wall and a component of 3 m s^{-1} perpendicular to the wall. The coefficient of restitution between the sphere and the wall is 0.5. Find the speed of the sphere immediately after the impact. [2]

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2. A smooth sphere moving on a smooth horizontal floor at 10 m s^{-1} strikes a smooth fixed vertical wall. [3]
Immediately before the impact, its direction of motion makes an angle of 60° with the wall. The coefficient of restitution between the sphere and the wall is 0.4.
Find the speed of the sphere immediately after the impact and the angle that its direction of motion then makes with the wall.

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3. A smooth sphere is moving on a smooth horizontal plane with velocity $(5\mathbf{i} - 4\mathbf{j})\text{ m s}^{-1}$ when it strikes [3]
a smooth fixed vertical wall which is parallel to $\mathbf{i}$. The coefficient of restitution between the sphere and the wall is 0.5.
Find the velocity of the sphere immediately after the impact.

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4. A smooth sphere of mass 0.5 kg is moving at 8 m s^{-1} when it strikes a smooth fixed vertical wall. [4]
Immediately before the impact, its direction of motion makes an angle of 30° with the wall. The coefficient of restitution between the sphere and the wall is 0.5.
Find the kinetic energy lost in the impact.

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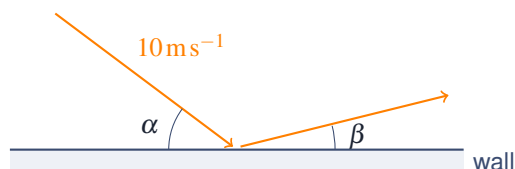
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5. A smooth sphere S is moving on a smooth horizontal plane with speed 10 ms^{-1} when it strikes a smooth fixed vertical wall. Immediately before the impact, the direction of motion of S makes an angle α with the wall, where $\tan \alpha = \frac{3}{4}$. Immediately after the impact, the direction of motion of S makes an angle β with the wall. The coefficient of restitution between S and the wall is $\frac{1}{3}$.



- (a) Show that $\tan \beta = \frac{1}{4}$. [3]

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- (b) Find the speed of S immediately after the impact. [2]

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- (c) Find the angle through which the direction of motion of S is deflected by the impact. [2]

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- (d) Find the fraction of the kinetic energy of S that is lost in the impact. [2]

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(Total for Question 5 is 9 marks)

**Section B Oblique impact of two smooth spheres**

21 marks

6. Two smooth spheres A and B collide. At the instant of impact the line of centres is parallel to $\mathbf{i}$. [2]
Immediately before the impact, A has velocity $(3\mathbf{i} + 4\mathbf{j}) \text{ m s}^{-1}$. Immediately after the impact, the $\mathbf{i}$ -component of the velocity of A is -1 m s^{-1} .
Write down the velocity of A immediately after the impact and find its speed.

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7. Two smooth spheres A and B , of equal radius and masses 2 kg and 1 kg respectively, are moving on a smooth horizontal plane. A has velocity $(4\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$ and B has velocity $(-2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when they collide. At the instant of collision the line of centres is parallel to $\mathbf{i}$. The coefficient of restitution between A and B is 0.5 . [3]
Find the velocities of A and B immediately after the collision.

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8. A smooth sphere A moving with speed 6 m s^{-1} collides with an identical smooth sphere B which is at rest. Immediately before the impact, the direction of motion of A makes an angle of 30° with the line of centres. The collision is perfectly elastic. [3]
Find the speed and direction of motion of each sphere immediately after the collision.

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9. A smooth sphere A of mass m moving with speed u collides with a smooth sphere B of mass m and equal radius, which is at rest. Immediately before the impact, the direction of motion of A makes an angle of 60° with the line of centres. The coefficient of restitution between the spheres is $\frac{1}{2}$. Show that the speed of A immediately after the collision is $\frac{7}{8}u$, and find the speed of B . [4]

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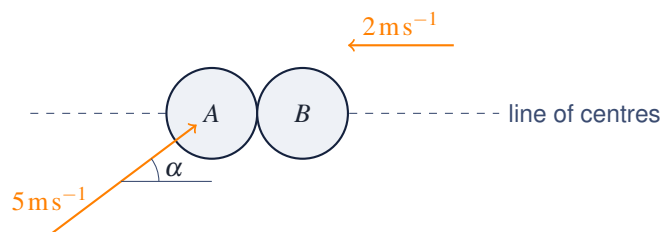
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10. Two smooth spheres A and B , of equal radius and masses 2 kg and 3 kg respectively, collide on a smooth horizontal plane. Immediately before the collision, A is moving with speed 5 ms^{-1} in a direction making an angle α with the line of centres, where $\tan \alpha = \frac{3}{4}$, and B is moving with speed 2 ms^{-1} along the line of centres towards A , as shown. The coefficient of restitution between A and B is $\frac{1}{2}$.



- (a) Find the components of the velocities of A and B along the line of centres immediately after the collision. [4]

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- (b) Find the speed of A immediately after the collision, and the angle its direction of motion then makes with the line of centres. [3]

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- (c) Find the kinetic energy lost in the collision. [2]

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(Total for Question 10 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

11. A smooth sphere A moving with speed u collides with an identical smooth sphere B which is at rest. [5]
Immediately before the impact, the direction of motion of A makes an angle of 30° with the line of centres. Immediately after the impact, the direction of motion of A makes an angle of 60° with the line of centres. The coefficient of restitution between the spheres is e .
Find e , and find the fraction of the total kinetic energy that is lost in the collision.

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END OF WORKSHEET

Total: 47 marks